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1 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
2 % Standard Parameters Xd,Xq,Xd1,Xd2,..Td01,Td02,..
3 % to Fundamental absolut parameters for ltspice model
4 %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
5
6 % Physical parameters of Synchronous generator
7 SN=250e6
8 UN_verkettet=13800
9 UN=UN_verkettet/sqrt(3)
10 cosphi=0.8
11 Pel=SN*cosphi
12 IN=SN/UN_verkettet/sqrt(3)
13 ZN=UN/IN
14 ZBase=UN/IN
15 fN=60
16 omega=2*pi()*fN
17 % Exciter Voltage and current
18 Ue0=200 % (choosen free)
19 Ie0=767 % (choosen to meet Td01)
20 % Input of Standard Parameters(Reactances and time constants)
21 x1=0.18
22 xd=1.305
23 xq=0.474
24 xd1=0.296
25 xq1=0
26 xd2=0.252
27 xq2=0.243
28 rcu=0.003
29 % open circuit time constants
30 Td01=4.455 % Td0'
31 Tq01=1      % Tq0'
32 Td02=0.053 % Td0''
33 Tq02=0.1    % Tq0''
34 % computed short circuit time constants
35 Td1=Td01*xd1/xd % (Td')
36 Td2=Td02*xd2/xd1 % (Td'')
37 % Intermediate calculation
38 Xd=xd*ZBase
39 Xq=xq*ZBase
40 xmd=xd-x1
41 xmq=xq-x1
42 xe=xmd^2/(xd-xd1)
43 xrd=xmd^2.0/(xd1-xd2)*(1.0-xmd/xe)^2.0+xmd^2.0/xe

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44 xrq=xmq^2/(xq-xq2)
45 sigmae=1-xmd/xe
46 sigmaxd=1-xmd/xd
47 sigmard=1-xmd/xrd
48 % Computation of Fundamental parameters for LTSpice model
49 % Delete Formulas and overwrite with the values of
50 % Rs,Ls,RE,RDx,RDy,LDx,LDy,LE,R0,L0 below
51 Rs=rcu*ZBase % Rs
52 Ls=x1*ZBase/omega % Ls
53 Lmd=xmd*ZBase/omega % (Lhd3)
54 Lmq=xmq*ZBase/omega % (Lhq3)
55 Lfd=Lmd*sigmae % Le
56 Lkd=(xrd-xmd)*ZBase/omega % LDx
57 Lkq=(xrq-xmq)*ZBase/omega % LDy
58 Rkd=(xrd-xmd^2/xe)*ZBase/omega/Td02 % RDx
59 Rkq=(xrq*ZBase/omega/Tq02) % RDy
60 Re=Ue0/Ie0*2/3 % RE
61 Re0=Ue0/Ie0
62 g=Ue0/(UN*sqrt(2)*Re0/(Xd-omega*Ls))
63 Res=Re*g*g
64 R0=Rs % R0
65 L0=Ls % L0
66 Td01=Xd/Res/omega % Should be same as above
67 % If not change Ie0 to meet Td01
68
69

```